Java Backend Interview Notes

Page 1 - Java Fundamentals

Java is an object-oriented programming language. Key concepts include encapsulation, inheritance, polymorphism and abstraction. HashMap stores key-value pairs using hashing and provides average O(1) time complexity for get() and put(). Streams support filter, map, sorted and collect operations. Exception handling uses try, catch, finally and throw.

Page 2 - Spring Boot

Spring Boot simplifies application development with auto-configuration, starter dependencies and embedded servers. Constructor injection is preferred over field injection because it improves immutability and testability. Common annotations include @RestController, @Service, @Repository and @Configuration.

Page 3 - Hibernate & JPA

Hibernate is an ORM framework that maps Java objects to relational tables. Lazy loading fetches related entities only when accessed, while eager loading fetches them immediately. Cascade types help propagate persistence operations. Transactions ensure data consistency.

Page 4 - SQL & Database

Indexes improve query performance by reducing full table scans. A clustered index determines the physical order of rows, while a non-clustered index maintains a separate index structure. Normalization reduces redundancy, whereas denormalization improves read performance in some scenarios.

Page 5 - REST & Security

REST APIs use GET for reading, POST for creating, PUT for updating and DELETE for removing resources. JWT is a stateless authentication mechanism where a signed token is issued after login. HTTPS encrypts communication between client and server. Proper HTTP status codes improve API design.